LEARNING OBJECTIVES

PATHOGENESIS/PATHOPHYSIOLOGY (normal disease state)

1. Describe the anatomy of the pituitary gland at the gross and microscopic levels, including blood supply and the gland's clinically relevant relationships (i.e. to cranial base, hypothalamus and optic nerve) and be able to identify cells and tissues of the pituitary via microscopy ANAT - **Ayla P**
2. Compare and contrast the embryonic origins of both pituitary lobes. ANAT - **Ayla P**
3. Describe the structural and functional anatomy of the thyroid and parathyroid glands and identify cells and tissues via microscopy. ANAT JP Done
4. Explain the positive (feed-forward) and negative feedback regulation of anterior pituitary hormones and target-organ responses. PHYSIO - **Spencer**
5. Recall the regulation of PTH secretion and role in maintaining calcium homeostasis. PHYSIO/CM JP Done
6. Describe the regulation and role of the neuropeptides secreted by the posterior pituitary. PHYSIO **Ravi**
7. Describe the location, function, and signaling mechanisms of thyroid hormone receptors and parathyroid hormone receptors. BCM - **Tomiwa**
8. Compare and contrast the genetics, clinical features (including organs affected), and laboratory findings of multiple endocrine neoplasia (MEN), types 1, 2A, and 2B. PATH **Ravi**
9. Recall the histopathology of pituitary adenomas and discuss the clinical features, laboratory findings, and immunohistochemical features of the most common types of pituitary adenomas (prolactinoma, somatotroph adenoma, corticotroph adenoma, and nonfunctional adenoma). PATH **Connor**

EPIDEMIOLOGY (incidence, prevalence, risk factors)

1. Describe the prevalence of sellar/suprasellar tumors in children and adults. CM **Ravi**

CLINICAL PRESENTATION (signs, symptoms, history, exam)

1. Identify and interpret the history and physical exam findings associated with sellar/suprasellar tumors including bi-temporal hemianopia. CS - **Spencer**
2. Review components of the normal thyroid exam and identify the history and physical exam findings in hypothyroidism. CS Sophie Done
3. Compare and contrast the clinical presentations of pituitary hormone excess and pituitary hormone deficiency focusing on TSH, prolactin, and ACTH. CM Sophie Done

DIAGNOSIS (differential diagnosis, diagnostic tests and tools, clinical reasoning)

1. Outline the steps in evaluating sellar masses or masses that cause pituitary compression (physical exam, biochemical evaluation and imaging studies). CM. **Connor**
2. Given a set of laboratory values, distinguish between hypothalamic, pituitary or target organ dysfunction. CM- **Tomiwa**

TREATMENT AND OUTCOMES

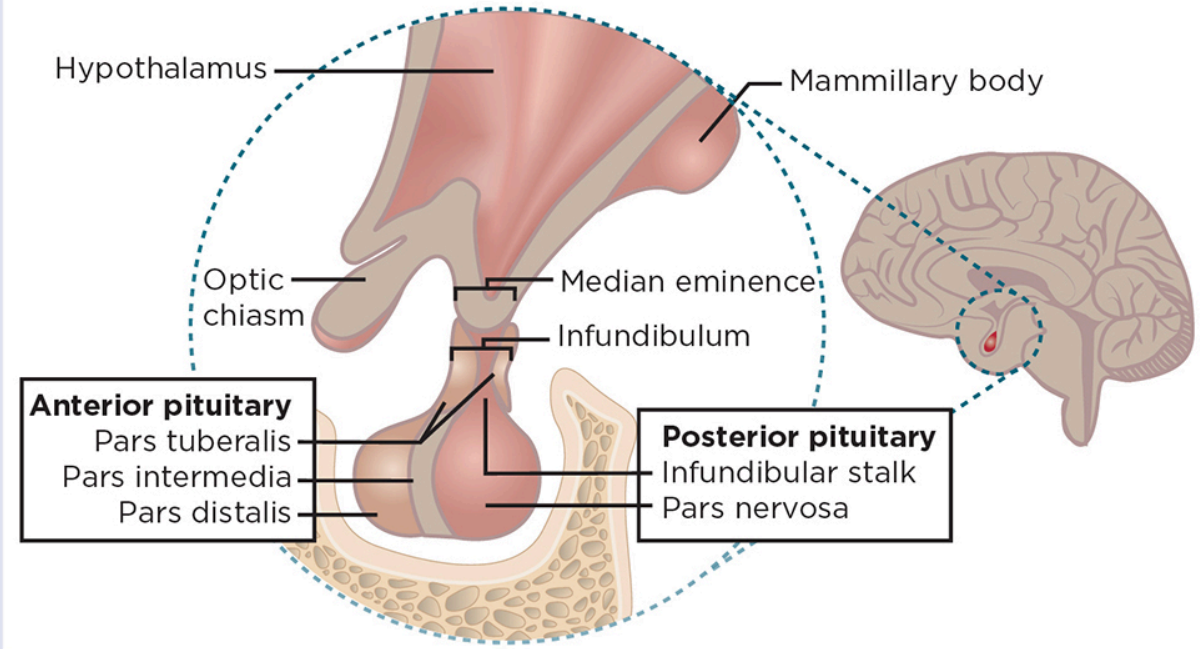
1. Describe the mechanism of action, adverse effects, and lab monitoring parameters for cabergoline when it is used in the medical treatment of prolactinomas. PHARM **Shannon (done)**

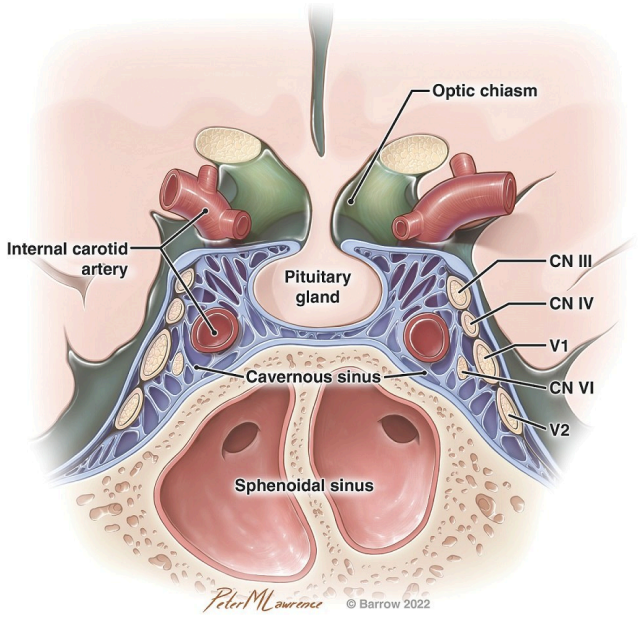
2. Explain the mechanism of action, adverse effects, and monitoring parameters of levothyroxine. PHARM - **Shannon (done)**
3. Describe the treatment of pituitary dysfunction and the management of sellar/suprasellar masses.CM **Shannon (done)**
4. Recognize the benefits of team-based care interactions organized around patient circumstances or conditions. HSS. (U of Texas Dell Medical School Module 4, Coordinating Care for Patients: http://vbhc.dellmed.utexas.edu//?id=m4/m4s1p1 (Note: create account to access this free module.) **Ishaan done**
5. Demonstrate ability to design care models coordinate around the needs of patients. HSS. (U of Texas Dell Medical School Module 4, Coordinating Care for Patients. http://vbhc.dellmed.utexas.edu//?id=m4/m4s1p1) **Ishaan done**

1. Describe the anatomy of the pituitary gland at the gross and microscopic levels, including blood supply and the gland's clinically relevant relationships (i.e. to cranial base, hypothalamus and optic nerve) and be able to identify cells and tissues of the pituitary via microscopy ANAT

Gross anatomy:

Fig 1. **Anatomy of the hypothalamus and pituitary gland**



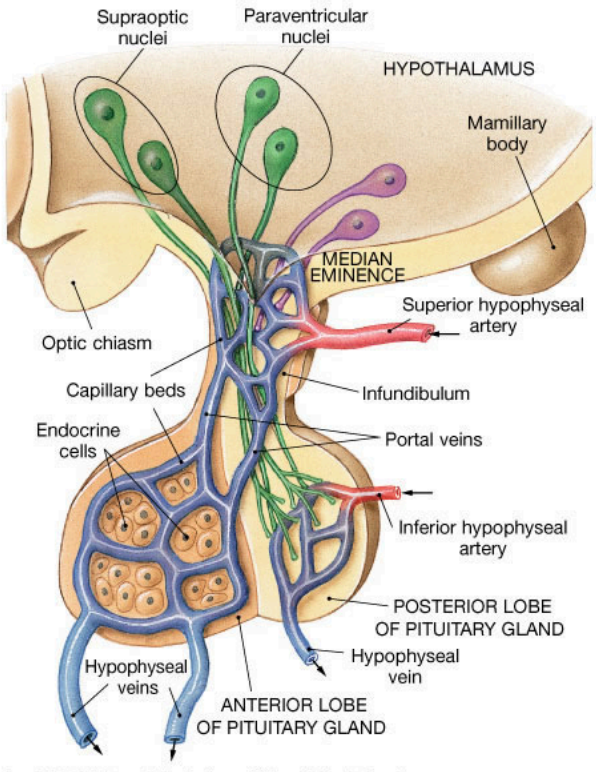


- Located in the sella turcica of the sphenoid bone

- Surrounded by numerous vital structures in the skull base: (“3 Cs”: carotid arteries, optic chiasm, cranial nerves III/IV/VI)
- Connected to the hypothalamus via the infundibulum, which contains the pituitary stalk (neurovascular connection from hypothalamus to posterior pituitary)
- Consists of two distinct regions: the anterior pituitary and posterior pituitary, which have different embryonic origins and blood supplies

Blood supply to the pituitary

- The anterior and posterior pituitary have separate blood supplies
- Posterior pituitary: supplied by the inferior hypophyseal artery (branch of the meningohypophyseal trunk of the ICA)
- Anterior pituitary: supplied by the hypophyseal portal system
    - Arterial supply: superior hypophyseal arteries (branches of ICA)
    - Primary capillary plexus: located in hypothalamus
    - Portal vein: travels down pituitary stalk
    - Second capillary plexus: within the anterior pituitary
    - Venous drainage: hypophyseal vein (drains to cavernous sinus)
- The purpose of portal circulation is direct delivery of hypothalamic regulatory proteins (releasing hormones) to the anterior pituitary



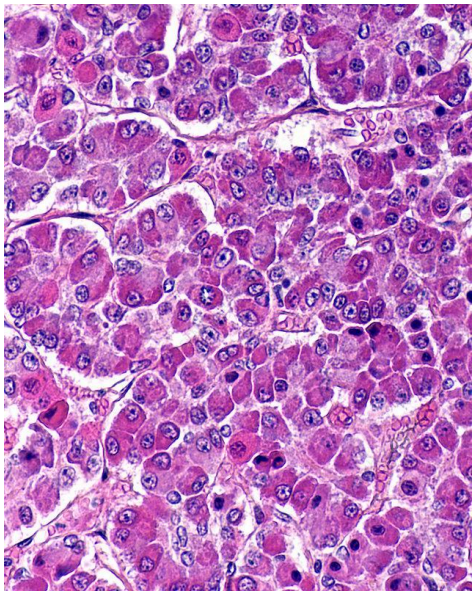
Microscopic anatomy

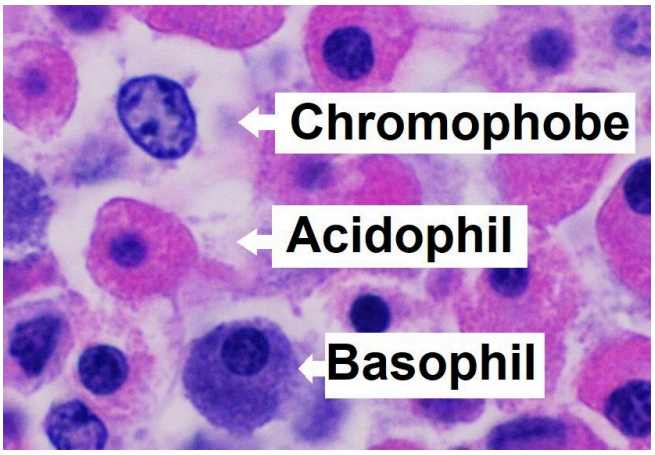
- The anterior pituitary is made up of three regions:

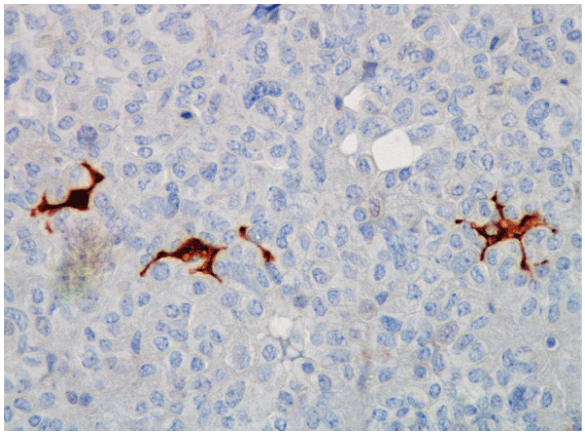
- Pars distalis: largest portion, responsible for most hormone secretion
- Pars intermedia: between pars distalis and posterior pituitary
- Pars tuberalis: wraps around infundibular stem

Histology of the pars distalis: 4 main cell types

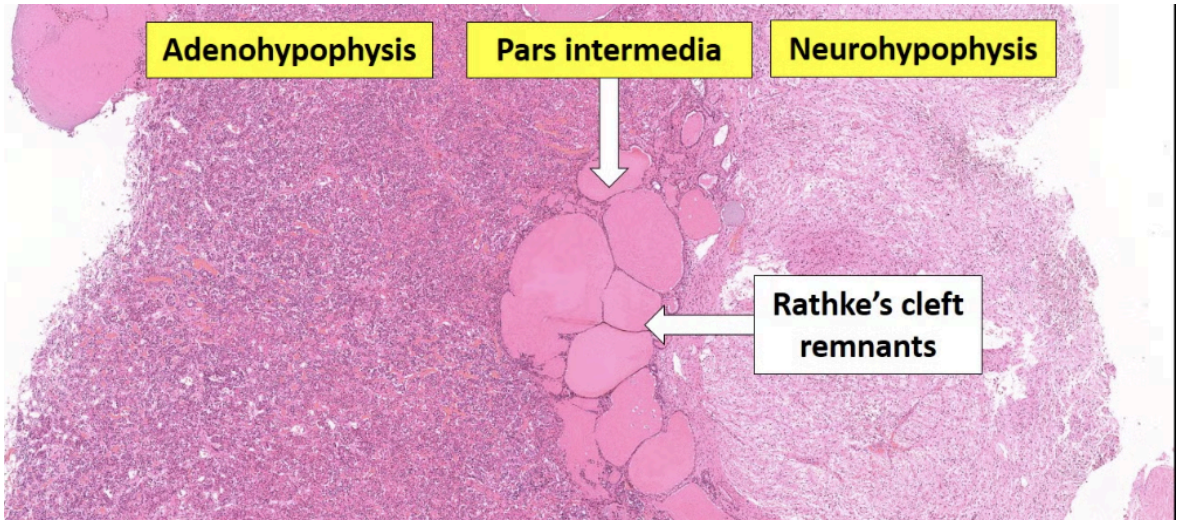
- Basophils: secrete FSH, LH, ACTH, TSH
- Acidophils: secrete prolactin and GH
- Chromophobes: mostly degranulated acidophils/basophils or undifferentiated stem cells; minimal activity
- Folliculostellate cells: star-shaped cells that secrete no hormones but provide support to basophils and acidophils; often visualized on IHC





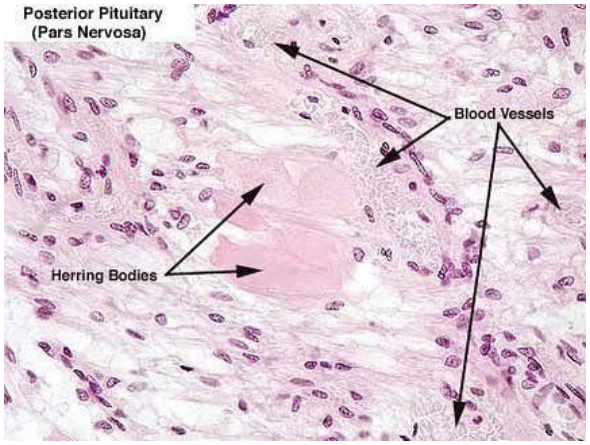


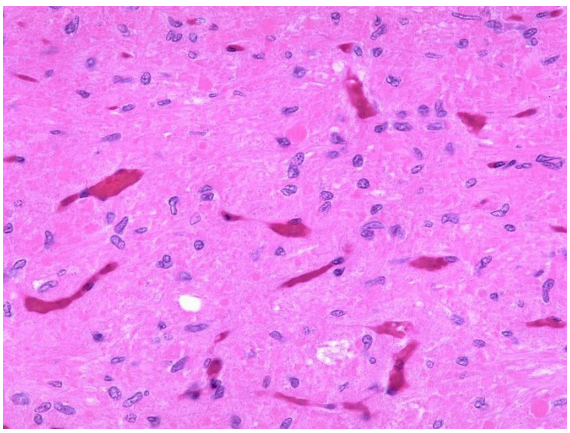
Histology of the pars reticularis and tuberalis: characterized by epithelioid clusters, colloid material may be visible in the remnants of Rathke's pouch



Histology of the posterior pituitary

- Pituicytes: glial-like cells
- Many capillaries
- Unmyelinated axons (axon terminals may appear as Herring bodies when full of ADH or oxytocin)





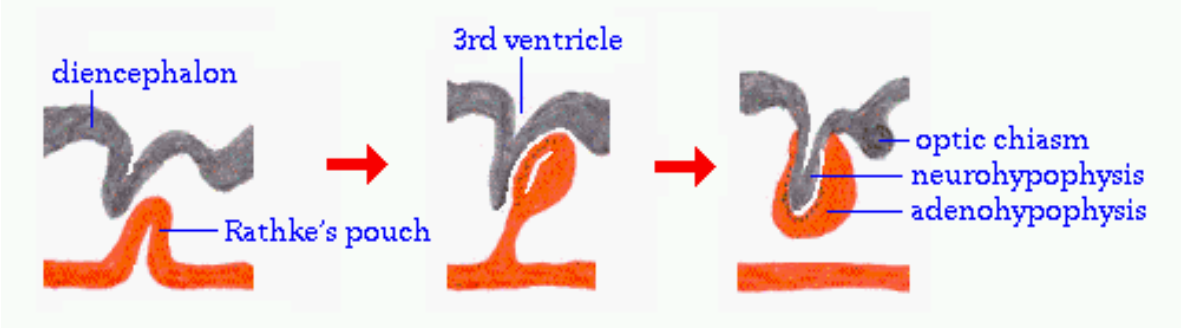
2. Compare and contrast the embryonic origins of both pituitary lobes. ANAT

Anterior pituitary (adenohypophysis): originates from oropharyngeal surface ectoderm

- Around week 3-4 of development, the oropharyngeal ectoderm evaginates upward, forming an area called Rathke's pouch which grows dorsally towards the brain. Eventually this pouch closes off, forming the pars disalis, pars intermedia, and pars tuberalis.
- Failure of Rathke's pouch to close completely results in a Rathke's cleft cyst, a collection of fluid between the anterior and posterior lobes of the pituitary. These cysts are generally asymptomatic but can cause mass effect if large enough.

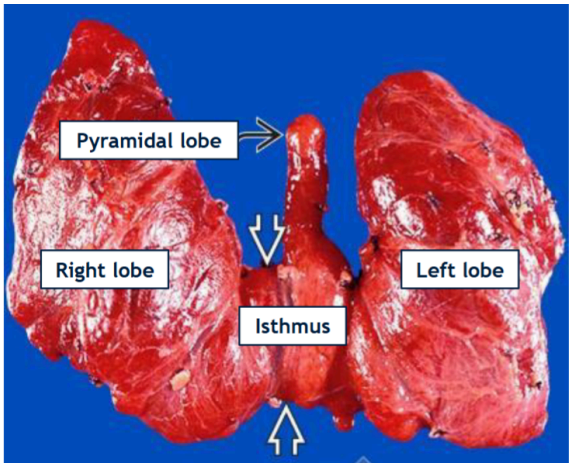
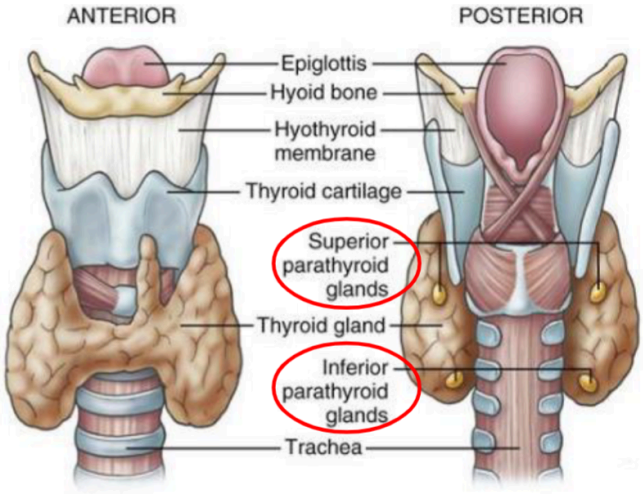
Posterior pituitary (neurohypophysis): originates from neuroectoderm at the base of the diencephalon, which grows toward Rathke's pouch to form the infundibulum and pars nervosa.

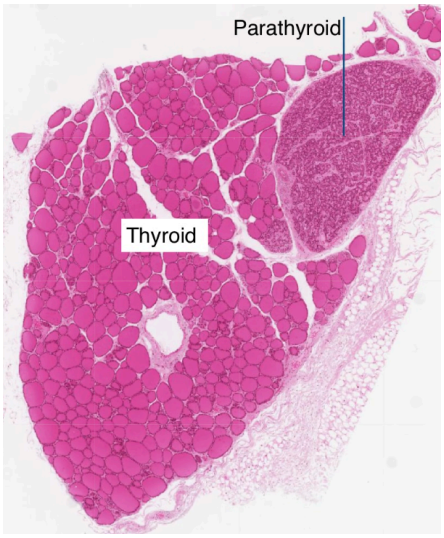
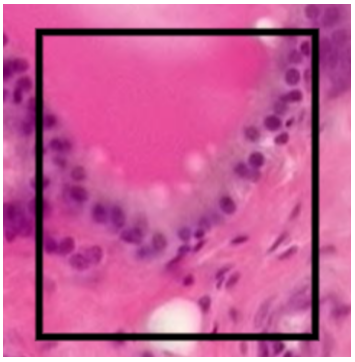
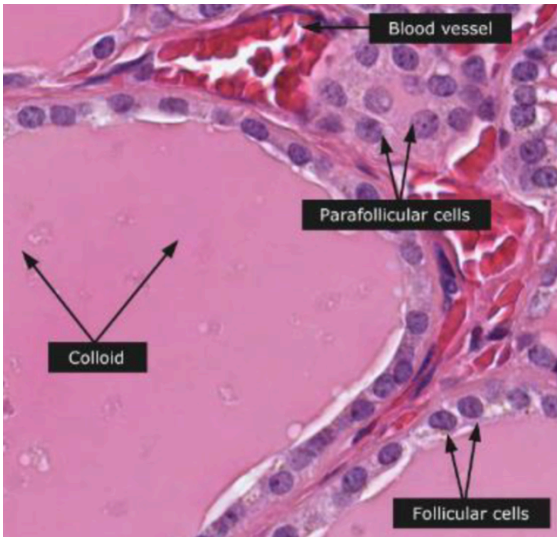
- Consists of neural tissue; extension of the hypothalamus
- Occurs during week 3-4 of gestation

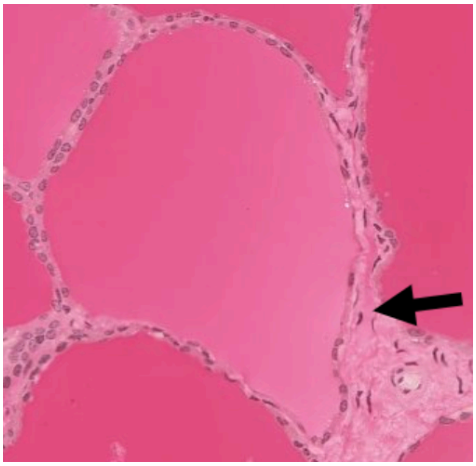
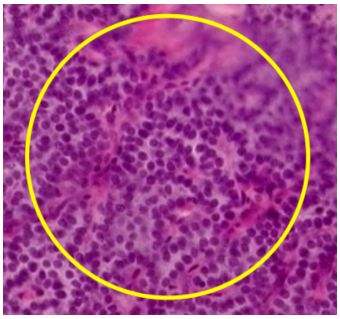
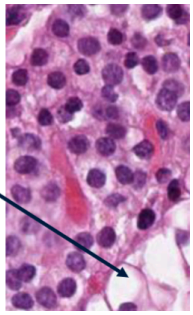
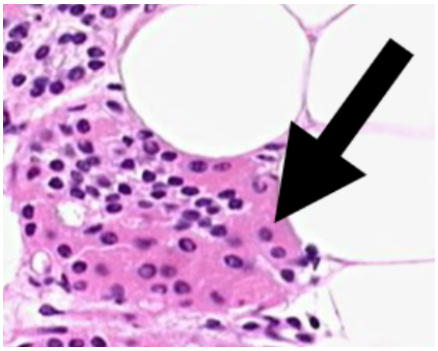


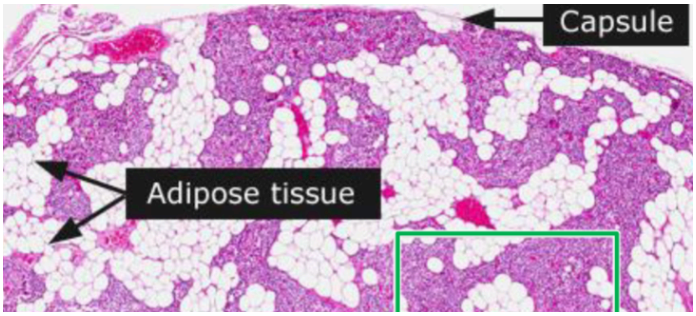
^visualization of how two separate tissues grow towards one another

1. Describe the structural and functional anatomy of the thyroid and parathyroid glands and identify cells and tissues via microscopy.

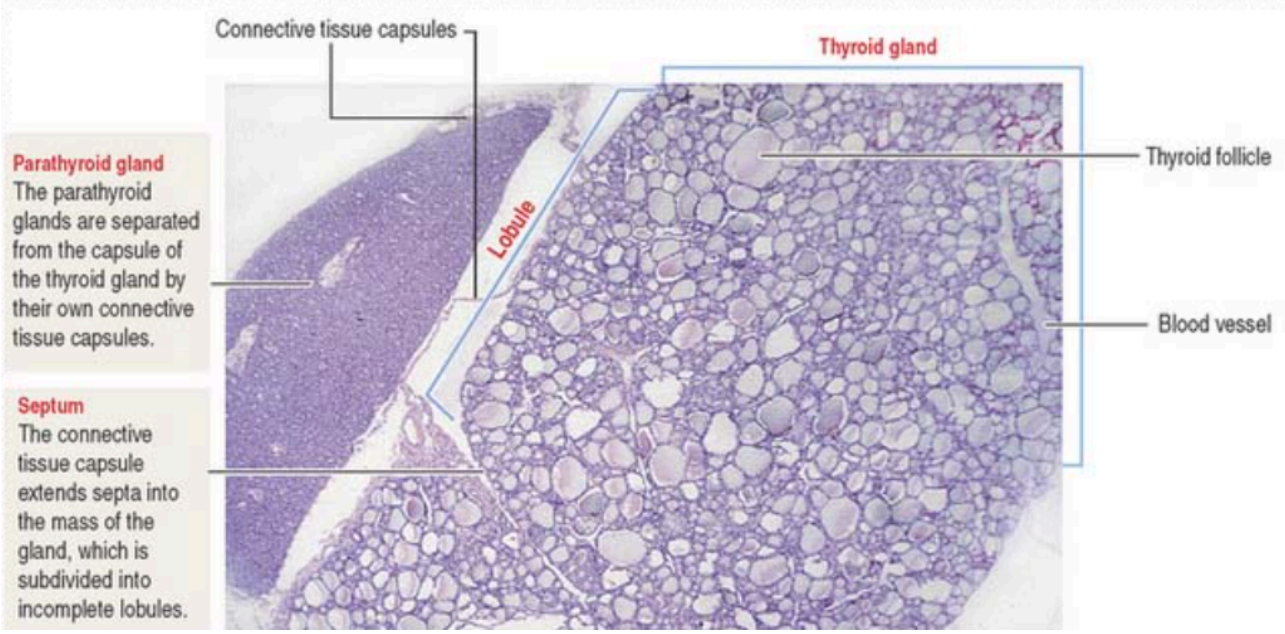
Name/Function	Image
Thyroid: bilobed endocrine gland located in the anterior neck consisting of two lateral lobes connected by a central tissue bridge (isthmus), occasionally featuring a superior projection known as the pyramidal lobe (~40% of cases) Thin fibrous capsule difficult to appreciate on gross image, histology at the end of this table!	
Parathyroid glands: typically four small, encapsulated nodular glands (paired superior and inferior) located on the posterior/dorsal aspect of the thyroid lobes, weighing 30–40 mg	

The thyroid gland contains follicular/principal cells, colloid, fenestrated capillaries, and parafollicular/C cells. The parathyroid gland contains chief/principal cells and oxyphil cells.	
Thyroid follicle: contains follicular epithelial cells around the periphery and a center filled with eosinophilic thyroid colloid. Active follicle has “scooped out” appearance (see black box) suggesting the cells are actively taking up colloid to release the thyroid hormones **Colloid:** homogeneous, eosinophilic substance occupying the follicular luminal cavity, composed mostly of thyroglobulin (large iodinated precursor glycoprotein that stores inactive thyroid hormones)	
Follicular cells (principal cells): simple epithelial cells lining the follicles that produce, store, and resorb thyroglobulin to release active T_3 and T_4. Morphology ranges from low cuboidal or squamous when inactive to tall columnar with apical endocytic vesicles when actively stimulated by TSH	

Fenestrated capillaries: surrounds thyroid follicles, facilitates rapid movement of hormones into the blood	
Parafollicular cells (C-cells): larger neuroendocrine cells with pale/clear cytoplasm located along the follicular basement membrane or in interstitial spaces between follicles; synthesize and secrete calcitonin to reduce blood calcium levels	 
Chief cells: Small, polygonal glandular epithelial cells arranged in cords/nests that primarily secrete Parathyroid Hormone (PTH)	 
Oxyphil cells: larger cells with small dark nuclei and abundant granular, eosinophilic cytoplasm densely packed with mitochondria. Can also secrete PTH hormone (unknown mechanism), express calcium sensors, and increase with age or uremic secondary hyperthyroidism	

Stromal adipose tissue (adipocytes): interstitial fat cells interspersed among parenchymal cells; normal adult tissue averages ~50% parenchymal cellularity to 50% fat (normal physiological range of 30% to 70% cellularity)	

Thyroid fibrous capsule: thin connective tissue covering that invaginates inward as projections (trabeculae) to subdivide the gland into poorly defined lobules



1. Explain the positive (feed-forward) and negative feedback regulation of anterior pituitary hormones and target-organ responses. PHYSIO

3-Tier Axis

Hypothalamus → Anterior Pituitary → Target Organ

- Feed-forward: Hypothalamic releasing hormones travel through the hypophyseal portal system and stimulate the pituitary.
- Trophic hormones: Pituitary hormones stimulate the peripheral target glands.
- Negative feedback: Target-gland hormones act back on the pituitary and hypothalamus to close the loop.
- Feed-forward signaling sets the pulsatile and circadian rhythms of secretion.

Feed-Forward (Stimulatory) Axes

- Adrenal (HPA): CRH → ACTH → Cortisol
    - AVP (vasopressin) also helps stimulate ACTH.
- Thyroid (HPT): TRH → TSH → T4 / T3
- Growth (Somatotropic): GHRH → GH → IGF-1 (made in the liver)
    - Ghrelin also boosts GH release.
- Gonadal: GnRH → LH / FSH → Sex steroids and gametes
- Prolactin (exception): Dopamine inhibits prolactin.

Prolactin exception: Prolactin is under tonic inhibition by dopamine. If the pituitary stalk is cut or the hypothalamus is damaged, prolactin goes UP and all the other pituitary hormones go DOWN.

True Positive Feedback

- Mid-cycle estradiol surge: When estradiol stays high, its effect flips from inhibiting to stimulating GnRH and LH, which produces the LH surge that causes ovulation.
- This is the main physiologic exception to the negative-feedback rule.

Negative Feedback Loops

1. Long-loop: The target hormone inhibits both the hypothalamus and the pituitary.

- Cortisol ⊣ CRH and ACTH
    - Acts through glucocorticoid and mineralocorticoid receptors, with both rapid (nongenomic) and slow (genomic) effects.
- T3/T4 ⊣ TRH and TSH
- Sex steroids ⊣ GnRH and LH/FSH

- IGF-1 ⊣ GH

2. Short-loop: The pituitary hormone inhibits the hypothalamus.

- GH inhibits its own secretion.

3. Ultrashort / Counter-regulatory: Local or opposing peptides act directly on the pituitary.

- Somatostatin opposes GHRH by lowering cAMP, which suppresses GH.

Clinical Application: Primary vs. Secondary Disease

Primary disease (the target organ fails):

- Pituitary hormone is ↑ HIGH because negative feedback is lost.
- Target hormone is ↓ LOW.
- Primary hypothyroidism: ↑ TSH, ↓ T4
- Addison disease: ↑ ACTH, ↓ cortisol

Secondary / Central disease (the pituitary or hypothalamus fails):

- Pituitary hormone is ↓ LOW or inappropriately normal.
- Target hormone is ↓ LOW.

Nuance

- Pulse generation is not purely top-down. Pituitary cells can produce oscillating output on their own, and peripheral factors act directly on the pituitary.
- The system is better thought of as circular than as a strict hierarchy.

Quick Review

1. Feed-forward: Hypothalamic releasing hormones drive pituitary trophic hormones, which drive target hormones. Prolactin is the exception because dopamine holds it down.
2. Negative feedback (long, short and ultrashort loops) keeps hormone levels stable. The estradiol-driven LH surge is the key example of positive feedback.
3. Primary disease: high pituitary hormone with low target hormone. Secondary disease: both are low or inappropriately normal.

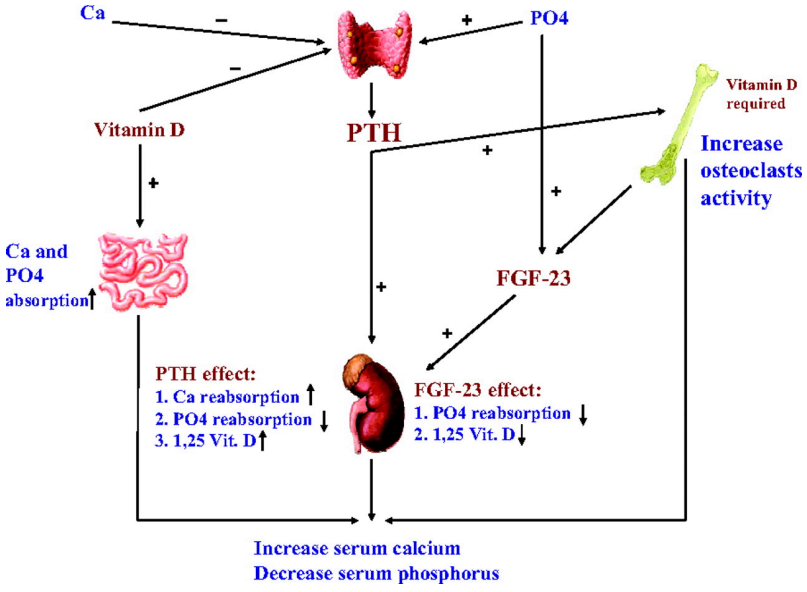
1. Recall the regulation of PTH secretion and role in maintaining calcium homeostasis. PH

PTH is released when the body detects a drop in calcium via **CaSR receptors** on parathyroid cells. It binds to osteoblasts which signal to **activate osteoclasts**, which causes an **increase in bone resorption** (breaks down bone tissue) which causes an increase in calcium and phosphate in the blood.

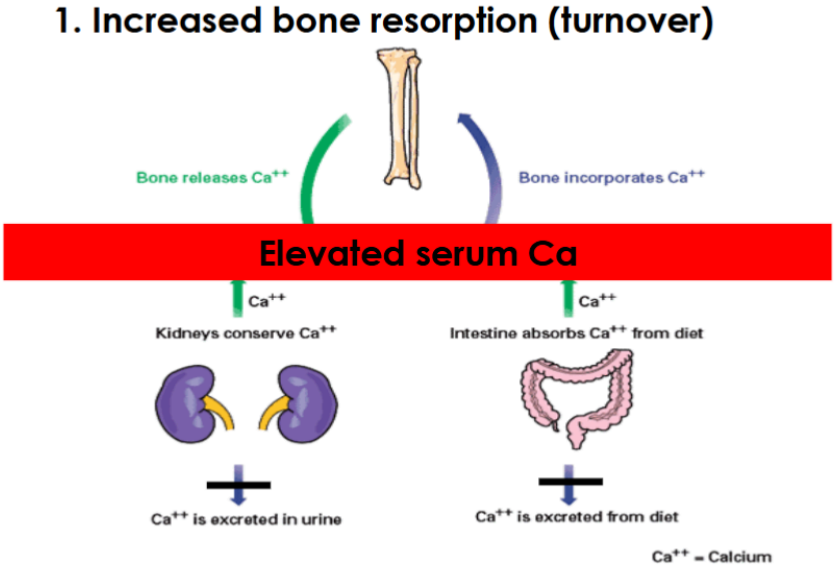
The kidneys respond to PTH by increasing calcium reabsorption in the DCT (primary site) and TAL (enhancing calcium movement). The kidneys also **decrease phosphate reabsorption** (leading to phosphaturia) and **increase 1α-hydroxylase activation** in the PCT which then **activates vitamin D** in the kidneys (pathway suppressed by high phosphate and FGF23).

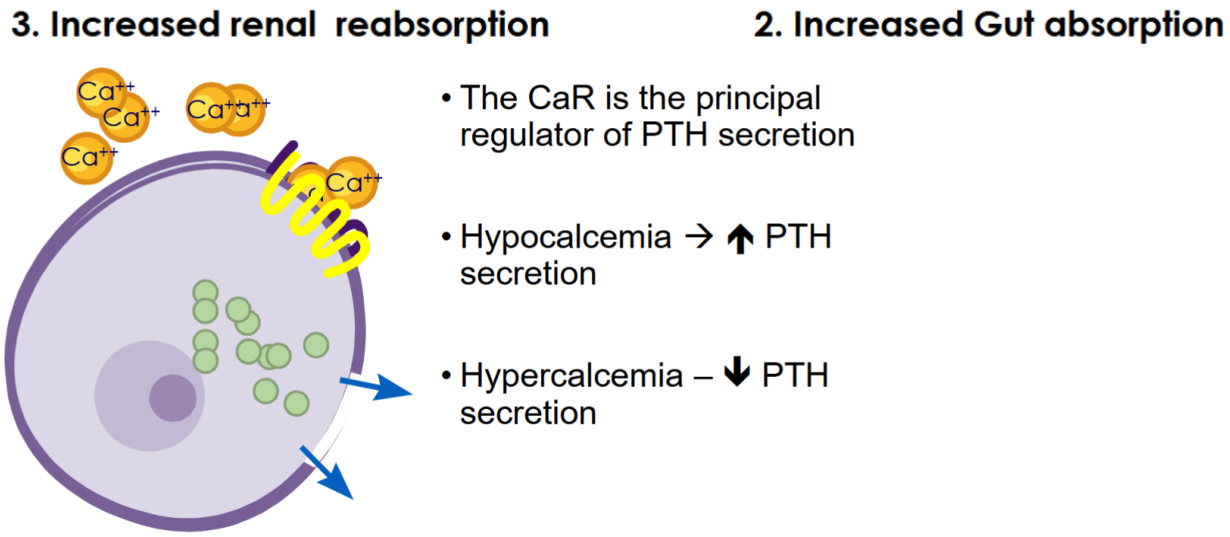
Vitamin D/calcitriol binds to the **nuclear VDR receptor** in the kidneys (to increase calcium and phosphate reabsorption) and in the gut (to upregulate **calbindin** expression, which upregulates the movement of calcium from the apical to basolateral membrane during intestinal absorption and also enhances intestinal phosphate absorption to nearly 80% efficiency with a less understood mechanism). Vitamin D/Calcitriol increase will have a **negative feedback loop** on PTH production. **Calcitonin decreases bone resorption** by inhibiting osteoclasts.

	$1,25\text{-}(OH)_2\text{-}D_3$	Parathyroid horomone	Calcitonin
Plasma $[Ca^{2+}]$	↑	↑	↓
Plasma $[PO_4]$	↑	↓	↓
Kidney	↑ Ca^{2+} reabsorption ↑ PO_4 reabsorption	↑ Ca^{2+} reabsorption ↓ PO_4 reabsorption ↑ 1α-hydroxylase activity	↓ Ca^{2+} reabsorption ↓ PO_4 reabsorption
Bone	promote PTH actions	↑ bone resorption	↓ bone resorption
GI Tract	↑ Ca^{2+}, PO_4 absorption	indirect by ↑ $1,25\text{-}(OH)_2\text{-}D_3$	none

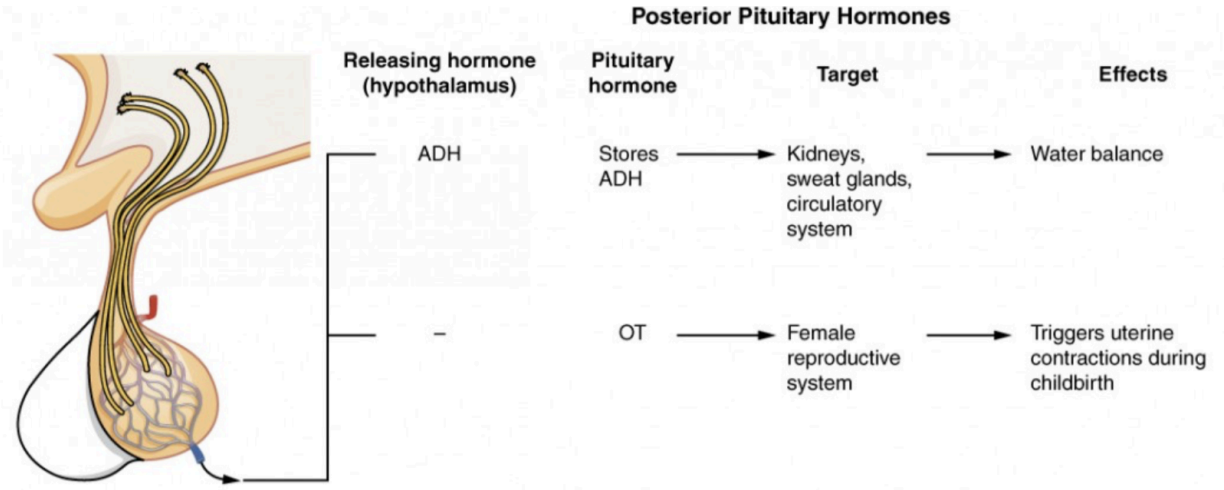


Helpful images from class today:





1. Describe the regulation and role of the neuropeptides secreted by the posterior pituitary. PHYSIO



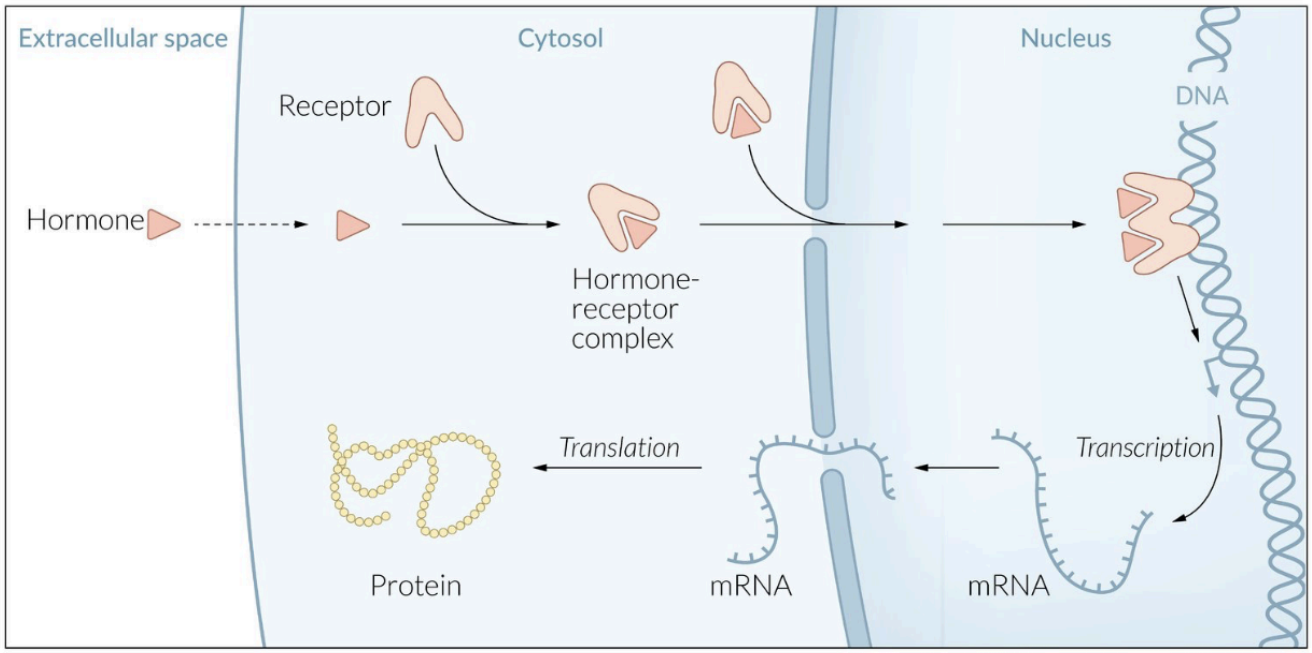
Made in	Supraoptic nucleus	Paraventricular nucleus
Released by	**High osmolality** (most sensitive) **Low volume / BP** (strongest)	**Suckling** **Cervical stretch** (positive feedback)
Blocked by	Alcohol, ANP	Stress, alcohol
Receptor	**V2** (kidney) / V1 (vessels)	Oxytocin receptor (Gq)
Main action	**AQP2** in collecting duct = water reabsorption	**Milk ejection** **Uterine contraction**
Too little	Diabetes insipidus (dilute urine, high Na)	Poor milk ejection
Too much	SIADH (concentrated urine, low Na)	Water intoxication (high-dose Pitocin)

1. Describe the location, function, and signaling mechanisms of thyroid hormone receptors and parathyroid hormone receptors

Thyroid Hormone Receptor	
Location	Type 2 NHR already associated w DNA in the nucleus
ligands	Mainly t3; T4 (will be converted intracellularly to T3)
Function	Regulates gene transcription involved in metabolism, thermogenesis, growth, nervous-system maturation, and β-adrenergic responsiveness
signaling	- TH-r become heterodimers w RXR (retinoid x receptor) and binds thyroid home response elements. - When T3 is not bound, TH-r recruits co-repressors that recruit histone deacetylase to condense chromatin and repress transcription - When T3 binds, a co-repressor is exchanged for a co-activator to drive histone acetylation, chromatin opening and transcription.

Parathyroid Hormone Receptor	
Location	On cell-surface PTH1 receptors, especially on osteoblast-lineage cells and renal tubular cells
ligands	Endocrine pTH and paracrine PTHrP
Function	Raises serum calcium and lowers serum phosphate ie Ca2+ and phos homeostasis for skeletal development and bone turnover - Decreases proximal tubular phos reabsorbption in kidneys - Increases distal tubular Ca reabsorption - Stimulates 1,25-dihydroxyvitamin D synthesis - Bone: mobilizes Ca and phos by increasing RANKL expression
signaling	This is a GPCR that is associated with a second messenger. PTH binds PTH1-r (GPCR) Two second messenger pathways involved: - Gαs → adenylyl cyclase → inc. cAMP → inc. PKA. predominant pathway in bones and kidneys - Also, Gq → phospholipase C → IP3/DAG → increased PKC and intracellular Ca2+

PTH secretion itself is regulated by CaSR on parathyroid chief cells: low extracellular calcium increases PTH secretion, whereas high calcium suppresses it. CaSR is a separate calcium-sensing GPCR, not the PTH receptor.


Extracellular space
Cytosol
Nucleus
Receptor
DNA
Hormone
Hormone-
receptor
complex
Translation
Transcription
Protein
mRNA
mRNA

1. Compare and contrast the genetics, clinical features (including organs affected), and laboratory findings of multiple endocrine neoplasia (MEN), types 1, 2A, and 2B.

	MEN 1 (Wermer)	**MEN 2A (Sipple)**	**MEN 2B (MEN 3)**
Gene	**MEN1** (menin), chr 11q13 Tumor suppressor, loss of function (two-hit)	**RET** proto-oncogene, chr 10q11.2 Gain of function RTK; codon **634**most common	**RET M918T** (codon 918) Highest risk; ~50% de novo
Inheritance	Autosomal dominant	Autosomal dominant	Autosomal dominant
Mnemonic	3 Ps	2 Ps	1 P
Parathyroid	**~90-95%**, multigland hyperplasia; usually first and most common feature	~20-30%, hyperplasia	**Absent**
Pancreas / GI	Pancreatic/duodenal NETs (~40-70%): **gastrinoma** (ZES, most common functional), insulinoma, nonfunctioning; foregut carcinoids	None	Intestinal **ganglioneuromatosis**(constipation, megacolon)
Pituitary	~30-40%; **prolactinoma** most common	None	None
Thyroid	None	**Medullary thyroid ca** (>95%), from C-cell hyperplasia	**MTC ~100%**, earliest and most aggressive
Adrenal	Adrenocortical adenomas (usually nonfunctioning)	**Pheochromocytoma** ~50%, often bilateral	**Pheochromocytoma** ~50%
Other features	Facial angiofibromas, collagenomas, lipomas	Cutaneous lichen amyloidosis; Hirschsprung disease	**Mucosal neuromas** (lips, tongue, eyelids), **marfanoid habitus**, thick lips, medullated corneal nerves
Key labs	High Ca + high/inappropriate PTH; fasting gastrin (secretin test: paradoxical rise); prolactin, IGF-1; insulin + C-peptide in hypoglycemia; chromogranin A	**Calcitonin**, CEA (MTC); plasma free / urine fractionated **metanephrines**; Ca, PTH; RET testing of relatives	Calcitonin, CEA; metanephrines; normal Ca/PTH; RET testing
Management hook	Subtotal (3.5 gland) parathyroidectomy; PPI for ZES	Prophylactic thyroidectomy by ~age 5 (codon 634)	Prophylactic thyroidectomy in **first year of life**

Recall the histopathology of pituitary adenomas and discuss the clinical features, laboratory findings, and immunohistochemical features of the most common types of pituitary adenomas (prolactinoma, somatotroph adenoma, corticotroph adenoma, and nonfunctional adenoma). PATH **Connor**

Subtype	Lineage / transcription factor	Hormone IHC	Clinical Features	Key Laboratory Findings
Prolactinoma	PIT-1	PRL-positive	**Amenorrhea, galactorrhea, infertility**, hypogonadism, low bone density	**Prolactin >250 µg/L**
Somatotroph	PIT-1	GH-positive	**Acromegaly/gigantism**, arthropathy, OSA, cardiomyopathy	**Increased IGF-1**, non-suppressible GH on OGTT
Corticotroph	T-PIT	ACTH-positive	**Cushing disease**; central obesity, rounded face, purple striae, easy bruising, proximal muscle weakness, hypertension	**Increased cortisol;** ACTH is elevated or inappropriately normal in pituitary-dependent disease
Nonfunctioning	Mostly SF-1 (gonadotroph)	FSH/LH mostly (depends on lineage)	Headache, visual field defects, fatigue, **reduced libido, menstrual irregularities**	No characteristic hormone excess; possible deficiencies of pituitary hormones

1. Describe the prevalence of sellar/suprasellar tumors in children and adults.

	Children	Adults
#1 tumor	**Craniopharyngioma** (cystic + calcified)	**Pituitary adenoma** (~90% of sellar masses)
Others	Optic glioma (NF1) Germinoma	Rathke cyst, meningioma Metastases
Adenomas?	**Rare** (~3%)	**Common** 1 in 1,000; 10-20% incidental
Presents with	Growth failure, DI, headache	Hormone excess, bitemporal hemianopia

Adenoma subtypes, most to least common

Prolactin > **Nonfunctioning** > GH > ACTH > TSH

1. Identify and interpret the history and physical exam findings associated with sellar/suprasellar tumors, including bi-temporal hemianopia. CS

- **Sellar tumors cause findings in two ways:**
    - Mass effect on nearby nerves and blood vessels
    - Hormone problems: either too much hormone (a functioning tumor) or too little (hypopituitarism from compression)
- Classic triad: headache, visual field loss and endocrine dysfunction
- **Bitemporal Hemianopia Mechanism:**
    - The tumor, most often a pituitary macroadenoma, grows upward and compresses the optic chiasm from below.
    - The fibers that cross at the chiasm come from the nasal retina, which sees the temporal visual fields.
    - Result: loss of both temporal visual fields.
- **Interpretation:**
    - Starts in the superior temporal quadrants, then progresses to complete bitemporal hemianopia.
    - Usually asymmetric between the two eyes.
    - Often comes with decreased visual acuity and impaired color vision.
    - Prognosis: Early compression is reversible. Long-standing compression causes permanent damage (optic disc atrophy), so longer symptom duration means worse visual recovery after surgery.
    - Junctional scotoma: central vision loss in one eye plus a superior temporal defect in the other eye. It localizes to where the optic nerve meets the chiasm.
    - Next step: formal visual field testing and an ophthalmology referral.
- **Other Mass-Effect Findings (By Direction of Growth)**
    - Upward (suprasellar): Chiasm compression. If the third ventricle is blocked, hydrocephalus follows, with vomiting, papilledema and decreased consciousness.
    - Sideways (cavernous sinus): Palsies of CN III, IV and VI, causing diplopia. CN III is the most vulnerable.
    - Stalk compression: Dopamine inhibition is lost, so prolactin rises. This suppresses GnRH and causes hypogonadism.
    - Hypothalamus: Diabetes insipidus, appetite and weight changes, temperature problems and behavior changes. These are more typical of craniopharyngioma.
    - Downward (sphenoid): CSF rhinorrhea
    - Headache: Common with all of these tumors, caused by stretching of the dura.

Hormone Excess (Functioning Tumors)

- **Prolactinoma (most common functioning tumor):**
    - Women: amenorrhea, galactorrhea, low libido, infertility
    - Men: erectile dysfunction, low libido, infertility
- **GH excess (Acromegaly / Gigantism):**
    - Coarse facial features; enlarged lips, nose, tongue, hands and feet
    - Rising hat, ring and shoe sizes; serial photos show gradual change

        - Deepening voice, sweating, joint pain, sleep apnea
        - Hypertension, diabetes, cardiomyopathy
        - Children (open growth plates): gigantism
- **ACTH excess (Cushing disease):**
    - Central obesity, hypertension, diabetes, osteoporosis
    - Depression, anxiety, insomnia
- **TSH excess:**
    - Hyperthyroid symptoms: weight loss, palpitations, sweating, irritability

Hormone Deficiency (Hypopituitarism from Compression)

- Hypogonadism is often the earliest sign in adults.
    - LH/FSH deficiency: low libido, erectile dysfunction, menstrual changes, fine wrinkled skin
    - TSH deficiency: fatigue, cold intolerance, weight gain, dry skin, constipation
    - TSH is low or inappropriately normal (central hypothyroidism)
    - GH deficiency: fatigue and weakness in adults; growth failure in children
    - ACTH deficiency (most dangerous): fatigue, hypotension, hypoglycemia, hyponatremia, poor stress tolerance

Special populations:

- Older adults: vague findings such as falls, confusion, drowsiness, hyponatremia and orthostatic hypotension
- Children (classically craniopharyngioma): growth failure, delayed puberty and signs of raised intracranial pressure

- Pituitary Apoplexy (Emergency)
    - Sudden bleeding into, or infarction of, a pituitary tumor, usually a nonfunctioning macroadenoma
    - Presentation:
        - Sudden "thunderclap" headache (about 80%)
        - Nausea and vomiting
        - Sudden vision loss
        - Ophthalmoplegia (CN III, IV or VI palsy) and diplopia
        - Meningismus and altered consciousness
    - Key interpretation:
        - Acute adrenal insufficiency can cause severe hypotension, so give IV glucocorticoids immediately.
        - Significant vision loss calls for urgent surgical decompression.

Quick Review

1. Bitemporal hemianopia = compression of the optic chiasm from below, which knocks out the crossing nasal fibers. It starts in the superior temporal fields.

2. Cavernous sinus invasion = palsies of CN III, IV and VI, with diplopia.
3. Stalk compression = mildly elevated prolactin, which leads to hypogonadism.
4. Hypogonadism is often the earliest hormone deficiency. ACTH deficiency is the most dangerous.
5. Thunderclap headache + vision loss + ophthalmoplegia = pituitary apoplexy. Give steroids immediately.

1. Review components of the normal thyroid exam and identify the history and physical exam findings in hypothyroidism. CS Sophie

Thyroid exam:

- Palpation: palpate the thyroid, should not be able to feel anything
- Thyroid is inferior to the cricothyroid membrane
- Even if it is **hypo**thyroidism, a goiter or a nodule may be felt

Lab exam:

- TSH, T3, T4, also thyroid antibodies

Short Thyroid Exam Video 1:38 long

	Hypothyroidism	Hyperthyroidism
Metabolism	Decreased	Increased
Weight	Gain, despite poor appetite	Loss, despite high appetite
Temperature	Cold intolerance	Heat intolerance
HR	Bradycardia	Tachycardia
GI	Constipation	Diarrhea
Skin	Cool, dry	Hot, sweaty
Hair	Coarse, thinning	Fine, thinning
Menstruation	Heavy and irregular	Light and irregular
Reflexes	Delayed relaxation	Hyperreflexia
Energy	Fatigue, slow thinking	Anxious, restless

Other features:

- Queen ann's sign: thinning of lateral portion of eyebrows
- Brittle nails
- Woltman sign: delayed reaction to reflexes
- a/w entrapment syndrome (carpal tunnel)
- Myxedematous heart disease

    - Dilated cardiomyopathy, bradycardia, dyspnea
- Non-pitting edema
- A/w hyperprolactinemia
    - Galactorrhea etc
- Depression

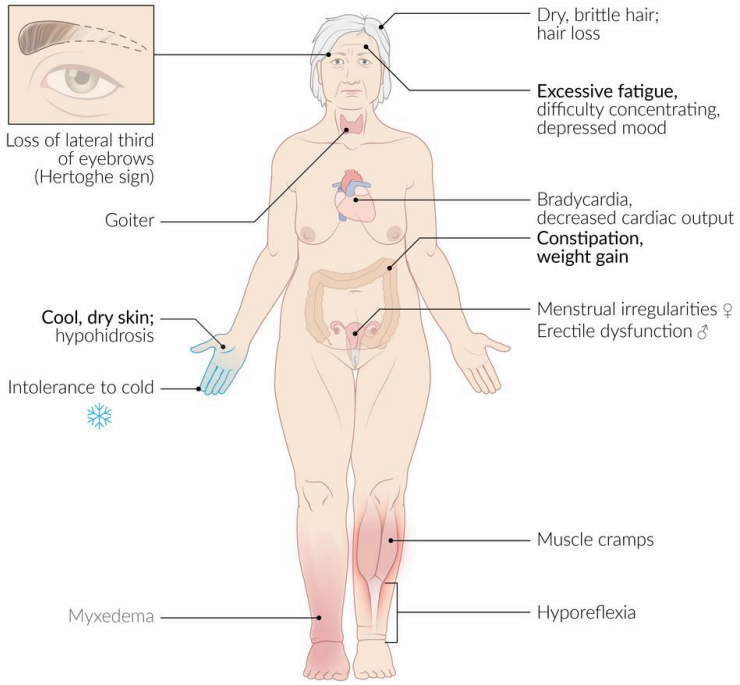
Hypothyroidism

Epidemiology
♀ > ♂ (4:1)
Prevalence: approx. 3% of the population
Incidence: increases with age (esp. after 60 years)

Etiology
Primary hypothyroidism: iodine deficiency (most common cause),
Hashimoto thyroiditis,
de Quervain thyroiditis,
iatrogenic (e.g., post-thyroidectomy)
Secondary hypothyroidism:
anterior pituitary insufficiency

Diagnosis
↑ TSH (primary hypothyroidism)
↓ TSH (secondary hypothyroidism)
↓ fT4

Complications
Myxedema coma



Thyroid Labs

	Etiologies	**TSH**	**FT4**	**FT3**
Primary Hypothyroidism	Hashimoto thyroiditis, iodine deficiency, thyroidectomy, radioactive iodine treatment, lithium, amiodarone	High	Low	Low/normal
Secondary Hypothyroidism (Pituitary)	Pituitary adenoma, pituitary surgery/radiation, Sheehan syndrome, pituitary apoplexy	Low/normal	Low	Low/normal
Tertiary Hypothyroidism (Hypothalamus)	Hypothalamic tumors, trauma, infiltrative disease	Low/normal	Low	Low/normal

Primary Hyperthyroidism	Graves disease, toxic multinodular goiter, toxic adenoma, iodine-induced hyperthyroidism	Low	High	High/normal
Secondary Hyperthyroidism (Pituitary)	TSH-secreting pituitary adenoma	High/normal	High	High

1. Compare and contrast the clinical presentations of pituitary hormone excess and pituitary hormone deficiency focusing on TSH, prolactin, and ACTH. CM Sophie

Clinical Presentations

	Hypopituitarism	Hyperpituitarism
TSH	Central Hypothyroidism: ● weight gain ● Fatigue ● cold intolerance ● bradycardia constipation (see LO C2)	Central Hyperthyroidism: ● weight loss ● Tachycardia ● heat intolerance ● Tremor ● sweating (see LO C2)
ACTH	Secondary Adrenal insufficiency: ● Fatigue ● Weakness ● Hypotension ● Hypoglycemia ● weight loss ● Cosyntropin does not elicit increase in aldosterone ● Low cort and aldo	Cushing syndrome: ● central obesity ● moon facies ● buffalo hump ● purple striae ● proximal muscle weakness ● Hypertension ● Hyperglycemia ● Hypokalemia ● Myopathy ● Psychosos ● Acne, hirsutism ● Menstrual issues ● Edema - *If plasma ACTH >20pg/mL likely ectopic or pituitary etiology* - *Cushing Disease (pituitary, 80% of time)* - *Ectopic ACTH secreting tumo 20% of time* - *If plasma ACTH <5pg/mL adrenal etiology* - *If plasma ACTH b/w 5-20 not definitive*

Prolactin	**Sheehan sd** • Failure to lactate in postpartum women	• Galactorrhea • Amenorrhea • Infertility • decreased libido • erectile dysfunction
GH	**Children**: dwarfism • short stature • failure to grow • Mild obesity • Delayed puberty **Adults**: • decreased muscle • increased fat • fatigue	**Children**: gigantism • Increased growth • Insulin resistance • Mild carb intolerance • Bitemporal hemianopsia **Adults**: acromegaly • Bones grow laterally, after the epiphyseal plates fuse • Coarse facial features • Wide fingers, hands, feet • Enlarged, muscular tongue • Sparse subcutaneous fat • Enlarged organs (heart) • Atherosclerosis • Hyperhydrosis • OSA • Colonic polyps • DM • Menstrual issues
LH/FSH	• Deficient testosterone/ estrogen • Delayed puberty • Amennorrhea • Erectile dysfunction • Decreased libido	• Rare to have a functioning gonadotroph adenoma that increase sex steroid levels **Hypogonadism** (loss of pulsatile secretion • Amenorrhea • Infertility • decreased libido • erectile dysfunction • delayed puberty

Outline the steps in evaluating sellar masses or masses that cause pituitary compression (physical exam, biochemical evaluation and imaging studies). CM. **Connor**

- Clinical symptoms
- Hormone studies (to confirm abnormal hormone secretion)
    - ACTH
        - 24-hour urinary free cortisol
        - Late-night salivary cortisol
        - Overnight dexamethasone suppression test
    - TSH
        - Thyroid function tests: TSH, T3/T4 levels
    - FSH/LH
        - LH/FSH levels
        - Testosterone/estrogen levels
    - Growth hormone
        - IGF-1 levels
        - Glucose tolerance test
    - Prolactin
        - Prolactin levels
- Imaging studies (to confirm source of abnormal hormone secretion)
- **MRI sella with IV contrast** (gold-standard)

1. Given a set of laboratory values, distinguish between hypothalamic, pituitary or target organ dysfunction. CM- Tomiwa

First, identify abnormal hormone and its controlling pituitary trophic hormone

Hypothalamus → pituitary → target organ hormone

- Target-organ dysfunction (primary): target hormone is abnormal, and the pituitary trophic hormone changes appropriately in the opposite direction because feedback is intact.
- Pituitary dysfunction (secondary): target hormone is low, with a low or inappropriately normal pituitary hormone.
- Hypothalamic dysfunction (tertiary): target hormone and pituitary hormone are low; hypothalamic hormone is deficient. Distinguishing hypothalamic from pituitary disease often requires a stimulation test.

Axis	Primary: target organ	Secondary: pituitary	Tertiary: hypothalamus
Thyroid	↑ TSH, ↓ free T4	↓/normal TSH, ↓ free T4	↓ TSH, ↓ free T4
Adrenal	↓ cortisol, ↑ ACTH	↓ cortisol, ↓/normal ACTH	↓ cortisol, ↓/ normal ACTH
Gonads	↓ testosterone/estradiol, ↑ LH/FSH	↓ testosterone/estradiol, ↓/ normal LH/FSH	Same pattern as secondary
GH axis	↓ IGF-1, usually ↑ GH if GH resistance	↓ IGF-1, ↓ GH	↓ IGF-1, ↓ GH

- For GH, interpret IGF-1, not a random GH level, because GH secretion is pulsatile. Low IGF-1 supports GH deficiency, but confirmation requires a GH stimulation test.
- A classic implication is that a low target-organ hormone with a normal trophic hormone is still central (secondary) disease if the trophic hormone is inappropriately normal, such as low free T4 with a “normal” TSH in central hypothyroidism.

Extra

Pitfalls that mimic central disease:

- Nonthyroidal illness (sick euthyroid): low T3/T4 with low/normal TSH in an acutely ill patient mimics central hypothyroidism — interpret thyroid axis cautiously during acute illness.
- Assay interference: heterophilic antibodies can spuriously alter TSH; confirm discordant free T4/TSH before diagnosing central hypothyroidism.
- Hormone resistance: high T3/T4 with inappropriately normal/elevated TSH suggests thyroid hormone resistance rather than a simple axis lesion.
- Masked deficits: cortisol deficiency can mask central diabetes insipidus until glucocorticoid replacement is started, so reassess the water axis after steroids.

Describe the mechanism of action, adverse effects, and lab monitoring parameters for cabergoline when it is used in the medical treatment of prolactinomas. PHARM

- Cabergoline = **long-acting ergot-derived dopamine D_2-receptor agonist**
    - Highly selective for D2 receptor
    - Primarily for hyperprolactinemia and prolactinomas
    - Physio TLDR: Does 2 things
        - 1. Suppresses prolactin release from lactotrophs → improving hyperprolactinemic symptoms
        - 2. Result = Antitumor effect = shrinking the adenoma.
    - Preferred over bromocriptine → more effective at lowering prolactin, shrinking tumors, and causes fewer AEs, also need less frequent dosing
- Physio
    - DA from hypothalamus travels down the portal circulation and acts on D2 receptors on pituitary lactotrophs → keeps prolactin release under tonic inhibition
    - Cabergoline mimics dopamine at these receptors
        - D2 is coupled to Gi/o → activation inhibits adenylyl cyclase → reduces Ca2+ influx
        - Reduced prolactin gene transcription and prolactin secretion
    - Eventually = Antitumor effect: lactotroph cell volume decreases w/ less cytoplasm, rough ER, and Golgi
        - Antiproliferative
        - Perivascular fibrosis
- Pharmokinetics
    - Half life = 65 hours (1-2x daily)
    - 0.25-0.5 per day, no more than 2mg per week
    - **HEPATICALLY/LIVER** metabolized (CYP3A4); excreted mainly in bile
    - Off-target activity: 5-HT2B receptor agonist (basis of valvulopathy and fibrosis risk)
- Adverse effects (bolded main ones)
    - **GI:** N/V, ABD discomfort, constipation
        - Take at bedtime with food and titrate slowly to mitigate
    - **Orthostatic hypotension**, dizziness, lightheadedness
    - **Somnolence/fatique** or **sleep attacks**
    - **Cardiac valvulopathy =** mitral, aortic or tricuspid <u>regurgitation</u> with leaflet thickening.
        - At prolactinoma doses, clinically significant valve disease is **RARE**, but a risk with cumulative dose is not excluded.
        - Cause: 5HT2B agonism
    - Pleural, pulmonary, pericardial and retroperitoneal **Fibrosis** are rare.
        - Cause: 5HT2B agonism
    - Headache
    - Nasal congestion
    - **Hallucinations, psychosis, confusion**
        - Caution with psychiatric illness
    - Depression, anxiety, mood changes

    - **Impulse-control disorders**, such as pathological gambling, hypersexuality, or compulsive shopping
    - Rarely, dyskinesia
    - With abrupt discontinuation: **Dopamine-agonist withdrawal syndrome**:
        - hyperthermia, rigidity, akinesia, reduced alertness, and increased creatine kinase, transaminases, and leukocytes.
        - Need to taper/withdrawl slowly → 2 years of therapy with normal prolactin + no tumor
            - Also need to monitor prolactin every 3 mo during first year of tapering, and annually after that (recurrence COMMON in first year)
- Other complications
    - Tumor-related (rapid shrinking) →
        - **CSF Rhinorrhea**
            - Can occur when macroadenoma has eroded the skull base and shrinks; meningitis risk
        - **Pituitary apoplexy**
        - **Chiasmal herniation** into an emptied sella → visual field loss
    - Vasospasm (**Raynaud's phenomenon**, digital vasospasms), peripheral edema, erythromelalgia
    - Suppression of lactation
    - Fertility → unplanned pregnancy can follow once prolactin normalizes
    - Resistance → failure to normalize prolactin or shrink tumor in maximally tolerated doses
        - Promptly checking adherence and then considering higher doses
- Contraindications (think 4 H's)
    - History of valvular **Heart disease**
    - Fibrotic disorders **History**
    - Uncontrolled **Hypertension**
    - preeclampsia/postpartum **Hypertension**
    - Ergot **Hypersensitivity**
- Lab Monitoring (listed in order of importance)
    - **1. Serum prolactin:**
        - Need baseline
        - Then about **1 mo** after starting and after each dose change
        - Once normal → check every 3-6 mo, then less often
        - Considerations:
            - Macroadenomas very high levels = check for HOOK EFFECT using diluted sample
                - If prolactin elevated w few symptoms = consider macroprolactin
            - Asymptomatic prolactinoma: every 6–12 months
    - **2. Pituitary MRI:**
        - Done at initial diagnosis and then…

    - **Macroadenoma:** repeat about 3 mos after starting, sooner if symptoms change. **Microadenoma:** repeat at about 1 year, or earlier if prolactin rises or symptoms recur.
    - Repeat before considering stopping the drug.
    - Repeat if symptoms develop or prolactin rises → assesses tumor shrinkage or regrowth.
- **3. Visual fields:**
    - Baseline + follow-up (assess if visual symptoms develop or imaging suggests optic-chiasm compression)
    - Repeated if have visual changes
- **4. Echocardiography**
    - Baseline before starting + repeated based on dosing/new murmurs
    - Assesses for valvulopathy
- 5. Other Pituitary Axes
    - Morning cortisol (± ACTH stimulation)
    - TSH with free T4 (TSH alone is unreliable)
    - LH/FSH with testosterone or estradiol, and IGF-1
    - Assess at baseline and periodically, since hypopituitarism may persist or recover.
- 6. Pregnancy test
    - Indicated for those who can get pregnant → ovulation returns quickly.
        - Stop cabergoline once pregnancy is confirmed, particularly for microadenomas
        - Macroadenomas need closer monitoring with symptom and visual field checks during pregnancy.
- 7. Renal and liver Function testing
    - Baseline
    - Hepatically metabolized so especially need LFTs; caution with hepatic impairment
- 8. BP
    - Baseline + start of medication → risk of hypotension
- 9. Psychiatric screening
    - Impulse control behaviors + mood changes at every visit.
- 10. Bone Density
    - Prolonged hypogonadism from hyperprolactinemia.
- 11. Respiratory and abdominal symptoms
    - ESR, creatinine, chest imaging or abdominal imaging if fibrosis is suspected.

Explain the mechanism of action, adverse effects, and monitoring parameters of levothyroxine. PHARM

- Levothyroxine is **synthetic L-thyroxine (T4)**, chemically identical to the hormone the thyroid makes.
    - T4 itself has low activity (prohormone) → **converted to T3 (active)**
        - In peripheral tissues (liver, kidney, muscle, brain, pituitary), 5′-deiodinases (D1 and D2) remove one iodine to form **T3**
            - 80% of circulating T3 comes from this conversion.
    - T4 and T3 cross cell membranes through transporters such as MCT8 and OATP1C1.
    - T3 binds **thyroid hormone receptors (TRα, TRβ)** (sit on DNA at thyroid response elements as heterodimers with the retinoid X receptor)
        - Binding switches the receptor from repressing → **activating**:
            - **More Na^+/K^+-ATPase activity** → higher O_2 consumption, thermogenesis + basal metabolic rate.
            - **More cardiac β_1-adrenergic receptors**, α-myosin heavy chain and SERCA2, giving **stronger contraction and faster heart rate**.
        - Effects on carbohydrate, lipid and protein metabolism, including more LDL receptors, which lowers LDL.
        - Essential roles in CNS development and skeletal growth in children.
    - T4 is converted to T3 within the pituitary and hypothalamus → suppresses **TRH and TSH**. This is why TSH is the main monitoring marker.
- **Pharmacokinetics**
    - **Absorption:** 40–80% of an oral dose is absorbed, mainly jejunum and ileum.
        - Best on an empty stomach, **<u>30–60 minutes before breakfast or at bedtime</u>**.
    - **Protein binding:** More than 99% is bound, mostly to TBG, with some to transthyretin and albumin.
    - **Half-life:** About 7 days in euthyroid people, longer in hypothyroidism and shorter in hyperthyroidism. This allows once-daily dosing, and **steady state takes about 6 weeks**.
    - **Therapeutic index:** Narrow. Switching between brands or generics can change levels.
- **Dosing**
    - Full replacement in healthy younger adults is about **1.6 mcg/kg/day**.
    - Elderly/patients with CAD → start low dose titrate slowly.
    - IV dosing is used for myxedema coma.
    - Higher doses are used for TSH suppression in differentiated thyroid cancer.
- **<u>AE (bolded important ones)</u>**
    - Correct dose, levothyroxine = no side effects.
    - AEs are almost all due to **overreplacement (iatrogenic thyrotoxicosis)**.

AE Class	**Specifics**
Cardiovascular *This is why start low*	Palpitations **Tachycardia** **Arrhythmias/Afib** → older adults with a suppressed TSH

and go slow	Angina, MI or heart failure in patients with underlying coronary disease.
Skeletal	Reduced bone mineral density **Osteoporosis**/fractures with chronic TSH suppression (postmenopausal women)
Hypermetabolic + Neuro	**Heat intolerance, sweating, weight loss** and increased appetite. Tremor, nervousness, anxiety and insomnia. Diarrhea. Menstrual irregularities. Muscle weakness. **Worsen glycemic control + insulin** or oral agent requirements (consideration for diabetes)
Adrenal Crisis	Pts w/ untreated adrenal insufficiency (ex. panhypopituitarism or autoimmune polyglandular syndrome) → levothyroxine raises cortisol clearance → triggers crisis. Replace glucocorticoids first.
Hypersensitivity	Rare; for excipients or dyes
Black box warning: Obesity/weight loss	Large doses can cause serious or life-threatening toxicity, especially with sympathomimetics
Specific to children	**Pseudotumor cerebri**, **slipped capital femoral epiphysis and craniosynostosis or premature bone age**. **Transient partial hair loss** can occur early in treatment.

- **Key drug interactions:**
    - **Reduced absorption:** calcium, iron, aluminum or magnesium antacids, bile acid sequestrants, sucralfate, sevelamer, PPIs and H2 blockers, soy, high-fiber foods and coffee. Separate these by about 4 hours.
    - **Higher dose needed because TBG rises:** oral estrogen, pregnancy, tamoxifen.
    - **Higher dose needed because clearance rises:** rifampin, phenytoin, carbamazepine, phenobarbital, and tyrosine kinase inhibitors.
    - **Warfarin:** Its effect is **potentiated** because clotting factor catabolism increases. Monitor INR when levothyroxine is started or the dose changes.
    - **Others:** Digoxin levels may fall as the patient becomes euthyroid. Amiodarone and lithium affect thyroid function directly.
- Monitoring parameters

Parameter (KNOW THIS)	When / target (EXTRA)
TSH (main marker in primary hypothyroidism)	**6–8 weeks** after starting or after any dose/brand/formulation change, or a new interacting drug. Once stable, every 6–12 months, then annually. Target = reference range (0.5–4.5 mIU/L, 0.5–2.5 in younger pts). Accept higher targets in older adults (ex. 4–6 mIU/L if over 70–80).

Free T4	**Central (secondary or tertiary) hypothyroidism:** TSH is unreliable, so titrate free T4 to the mid-to-upper normal range. Useful early in treatment of severe hypothyroidism, when TSH lags behind. Draw blood **before** the daily dose, since free T4 rises transiently after
Thyroid cancer	TSH suppression targets are set by ATA recurrence risk (below 0.1 mIU/L in high-risk disease and 0.5–2 mIU/L in low-risk disease) Weigh these against cardiac and bone risks.
Pregnancy	Increase the dose by about 25–30% as soon as pregnancy is confirmed Check TSH (± free T4) **every 4 weeks through mid-pregnancy**, then at least once around 30 weeks. Aim for the trimester-specific range (generally below 2.5 mIU/L). Return to the prepregnancy dose after delivery + recheck at 6 weeks.
Neonates and children	In congenital hypothyroidism, normalize free T4 within 2 weeks and TSH within 4 weeks. Check frequently: every 1–2 months in infancy, every 3–4 months later in childhood. Monitor growth, development and bone age.
Cardiac	Heart rate, BP, symptoms of angina or palpitations. ECG if atrial fibrillation is suspected, especially in older or cardiac patients during titration.
Bone	DXA in postmenopausal women or anyone on long-term TSH suppression.
Clinical response	Symptoms (fatigue, cold intolerance, constipation), weight and signs of over- or under-replacement.
Related labs	INR on warfarin. Glucose or HbA1c in diabetes. Lipids may improve with treatment. Check cortisol before starting if adrenal insufficiency is suspected.

Common pitfalls:

- **Biotin** supplements can falsely alter TSH and free T4 immunoassays. Stop biotin about 2 days before testing.
- A persistently high TSH despite escalating doses suggests **poor adherence or malabsorption**: celiac disease, *H. pylori*, atrophic gastritis, interacting drugs, or taking the dose with food.
- After any dose change, wait the full 6 weeks before rechecking, because of the long half-life.

Describe the treatment of pituitary dysfunction and the management of sellar/suprasellar masses.CM

Pituitary dysfunction and sellar/suprasellar masses

1. General approach to a sellar mass

- Get the biochemical diagnosis **before imaging**.
    - About 10% of healthy people have incidental nonfunctioning tumors, so a mass on MRI means little until the labs are known.
    - **MRI is the most sensitive imaging test.**
        - **MRI sella w/ IV contrast**
        - **2nd line** = CT sella
            - When MRI contraindicated
            - Useful for bony anatomy and transsphenoidal surgical planning
- **Differential diagnosis:**
    - Pituitary adenoma (about 90% of sellar tumors, almost always benign);
    - Hypophysitis
        - Including checkpoint-inhibitor hypophysitis
    - Parasellar tumors (glioma or meningioma)
    - Aneurysm or hemorrhage
    - Infiltrative disease (sarcoidosis)
    - Empty sella.
    - Craniopharyngioma
    - Rathke cleft cyst
    - Germ cell tumors
    - Metatases
- **Size:** a microadenoma is under 1 cm; a macroadenoma is over 1 cm.
- **Work-up of any adenoma:**
    - **Screen for hormone excess:**
        - prolactin, IGF-1, TSH with free T4, LH/FSH, and a Cushing screen.
    - **Macroadenomas only:** also check for **hypopituitarism** and get **formal visual field testing**, because the optic chiasm sits just above the sella.
- **Pitfalls when reading prolactin:**
    - **Stalk effect:**
        - a nonfunctioning macroadenoma can raise prolactin modestly (usually below about 100 ng/mL).
            - It compresses the stalk, which blocks DA from reaching normal lactotrophs, so prolactin rises modestly
            - A dopamine agonist will normalize the prolactin by acting on those normal cells, but the tumor won't shrink. It still needs surgery
            - Unlike true prolactinoma
    - **Hook effect:**
        - A giant prolactinoma can give a falsely low result
        - Ask the lab to dilute the sample.

Rule out an aneurysm before any surgery.

2. Treatment of hormone excess (functioning adenomas)

Tumor	First-line	Other options
Prolactinoma (most common secreting type)	**Medical:** DA agonists (**cabergoline**, bromocriptine). Shrink the tumor, restore menses and fertility, used even when visual fields abnormal.	Surgery if the drug fails or severe mass effect. Drugs can sometimes be stopped after 1–2 years. Pregnancy → stop the drug once pregnancy Macroadenoma → watch visual fields.
Acromegaly (GH)	**Transsphenoidal surgery**	**Somatostatin** analogs (octreotide, lanreotide, pasireotide); **Pegvisomant** =GH receptor antagonist (monitor liver enzymes; **DA agonists** Radiation as an adjunct. Goal: normal IGF-1 for age and sex.
Cushing disease (ACTH)	**Transsphenoidal surgery**	Radiation as an adjunct Medical options: **pasireotide** (lowers ACTH release), **ketoconazole** or **metyrapone** (block cortisol synthesis). Bilateral adrenalectomy, with the pituitary irradiated first (prevent **Nelson syndrome**).
TSH-secreting (very rare)	**Surgery** ± radiation	**Octreotide** if the patient cannot have surgery, or to make them euthyroid before it. Do not ablate the thyroid, because that can make the tumor grow.
Gonadotroph or nonfunctioning (null cell)	**Surgery** ± radiation, usually for a macroadenoma causing mass effect	—

3. Treatment of hypopituitarism (hormone deficiency)

- Order of loss is
  - **GH, FSH and LH first**
  - **TSH, ACTH and ADH last**.
  - The most common cause is a **macroadenoma**. Each missing axis is replaced:
- **1. ACTH deficiency (secondary adrenal insufficiency):** a **medical emergency**, give **glucocorticoids immediately**.
  - Dosing: **hydrocortisone about 20 mg/day**
    - (15 mg in the morning, 5 mg in the late afternoon), or prednisone 5 mg or dexamethasone 0.5 mg each morning.
  - No mineralocorticoid is needed, since renin–angiotensin still controls aldosterone.
  - **Give glucocorticoids before levothyroxine.**
    - Thyroid hormone speeds cortisol clearance and can trigger adrenal crisis.
  - **Sick-day rules:**
    - double or triple the dose during illness.
    - Patients should carry an emergency hydrocortisone injection, get stress dosing for surgery, and wear a medical alert ID.
- **2. TSH deficiency: levothyroxine**,

    - starting lower in older or cardiac patients. **Titrate to free T4, not TSH.**
- **3. LH/FSH deficiency: estrogen or testosterone**.
    - Fertility requires gonadotropins (hCG ± FSH, or pulsatile GnRH for hypothalamic causes).
- **4. GH deficiency:** recombinant GH by daily subcutaneous injection.
    - Monitor **IGF-1**, plus growth in children.
    - Side effects: edema, headache, arthralgias, carpal tunnel syndrome, worse glucose tolerance.
    - Active malignancy is a relative contraindication.
    - **Starting GH can unmask central hypothyroidism or adrenal insufficiency**, so recheck free T4 and cortisol.
- **5. ADH deficiency (AVP deficiency, central diabetes insipidus): desmopressin (DDAVP)**, a V2 agonist.
    - Use the lowest dose that controls symptoms, with the first dose at bedtime.
    - **Monitor sodium**, because the main risk is **hyponatremia**.
    - Starting glucocorticoids can unmask diabetes insipidus.
- **5. Prolactin deficiency:**
    - No replacement
    - The only sign is inability to lactate after delivery, which may be the first sign of **Sheehan syndrome**. In that case, test all axes.

4. Special situations

- **Pituitary apoplexy** (bleeding or infarction into an adenoma):
    - Presents acutely with headache, vision loss, ophthalmoplegia and AMS
    - Management: MRI, endocrine labs, and **surgical decompression** if needed.
    - **Give IV hydrocortisone 100 mg right away**, without waiting for lab results.
- **Lymphocytic hypophysitis:**
    - usually in late pregnancy or after delivery. Replace whichever hormones are deficient; diabetes insipidus is possible.
- **Empty sella:**
    - primary empty sella usually has normal function. Secondary empty sella (after surgery or radiation) should prompt checking for deficiencies.
- **After transsphenoidal surgery:**
    - Watch sodium: diabetes insipidus around days 1–2, SIADH and hyponatremia around days 5–10 (sometimes as a triphasic response).
    - Check morning cortisol and give steroid coverage as needed.
    - Reassess all axes at about 6 weeks, and get an MRI at about 3 months.
- **After radiation:**
    - new hypopituitarism can develop years later, so hormone testing continues annually for life.

5. Pituitary incidentaloma follow-up

- **Hypersecretion** → treat the specific tumor as above.
- **Compression of the chiasm → surgery**.
- **Nonfunctioning and no compression** → serial MRI:
    - **Macroadenoma:** repeat MRI at **6 months**.
    - **Microadenoma:** repeat at **1 year, then 2 years, then 5 years** if stable. Very small lesions (under 4 mm) may not need repeat imaging.

1. Recognize the benefits of team-based care interactions organized around patient circumstances or conditions. HSS. (U of Texas Dell Medical School Module 4, Coordinating Care for Patients: http://vbhc.dellmed.utexas.edu//?id=m4/m4s1p1 (Note: create account to access this free module.) **Ishaan**

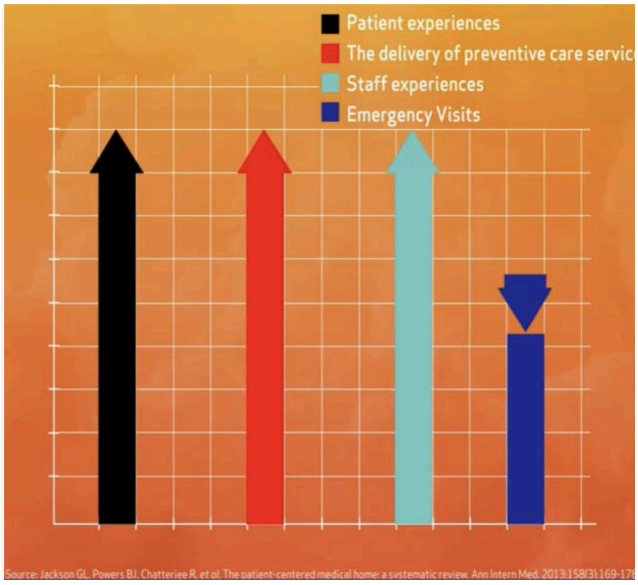
The problem: the manner in which our health care system is set up is conducive to inefficiencies and uncoordinated care.
The consequences: detriments to patient care → stress, serious harm, even death

- Wasted time
- Missed work
- Repeat testing
- Missed diagnosis
- Medical Errors

The solution: team-based care interactions intended to promote coordinated care!

Key elements of Teamwork

- Shared Goals: A team including providers, the patient, and where appropriate, people who support the patient, work together to establish goals that reflect what matters most to the patient
- Clear Roles: Each team member has clear expectations that allow them to take advantage of division of labor and accomplish more as a team than they would individually
- Mutual Trust: Team members earn each other's trust and share in achievements
- Effective Communication: There are consistent channels of communication that are prioritized and refined by all team members
- Measurable processes and outcomes: Performance on the team's shared goals is tracked over time and the team receives timely feedback on performance



Demonstrated benefits of team-based care

- Improved pt care
- Increased delivery of preventative care
- Improved staff experience
- Decreased emergency visits

1. Demonstrate ability to design care models coordinate around the needs of patients. HSS. (U of Texas Dell Medical School Module 4, Coordinating Care for Patients. http://vbhc.dellmed.utexas.edu//?id=m4/m4s1p1) **Ishaan'**

Patient Centered Medical Homes (PCMHs)

Patient Centered Medical Homes are a centralized source of primary care (+/- some specialty care) that promotes coordination

Components of a PCMH

- Co-located
- Unified database/EMR for patient information (can be accessed by all relevant providers easily)
- Care givers often know you personally and understand your full PMH
- Same-day appointments made available for patients
- Alternative methods of communication (such as telephone visits, emails, text messages)
- A systems-based approach to quality and safety
- Proactive visit planning initiated by the care team, not patients
    - E.g. A1c check for diabetes if you have not been seen for a while

PCMHs are ideal for …

- **Effective primary care**
- General Preventative Health & Wellness
- Multi-Morbidity Management
- Management of a chronic disease that requires follow up

Integrated Practice Unit (IPUs)

Integrated Practice Units are teams of both clinical and nonclinical members organized around a specific patient condition, who treat the conditions over the full cycle of care in a coordinated manner

Components of an IPU

- Co-located
- Organized around a medical condition
- Care for a patient over a defined "care cycle"/episode of care
- Includes several clinicians, case managers, social workers, with a physician team captain
- Measure outcomes, costs, and processes to improve as a unit
- *A challenge: reimbursement models that support IPUs*

IPU are ideal for

- Managing a specific disease or condition that requires many different clinicians

Whereas PCMHs are for generalized care of all patients, IPUs develop solutions for patients who share a condition or set of circumstances

While PCMHs provide longitudinal care over a patient's lifetime and generally regardless of his or her condition, IPUs tend to concentrate on conditions for which the care cycle is well-defined.